

What is Regression?

Regression is a statistical and machine learning technique used to **predict a continuous value** (a number) based on one or more input variables.

👉 In simple words:

It finds the relationship between **input (independent variables)** and **output (dependent variable)**.

Example:

- Predicting **house price** based on size, location, rooms
- Predicting **salary** based on experience

This falls under the field of Machine Learning and is based on the idea of modeling relationships between variables.

]

Common Types of Regression (with Examples)

1. Linear Regression

- Models a **straight-line relationship** between variables
- Equation: $y = mx + c$

Example:

- Predict salary based on years of experience
- If experience increases → salary increases linearly

2. Multiple Linear Regression

- Uses **more than one input variable**

Example:

- House price prediction based on:
 - Size
 - Location
 - Number of bedrooms

3. Polynomial Regression

- Models **curved relationships** instead of straight lines

Example:

- Predicting growth of a plant over time (not always linear)
- Sales trends that rise and fall

4. Logistic Regression

- Used when output is **categorical (yes/no, 0/1)**
- Despite the name, it's mainly used for classification

Example:

- Will a student pass or fail?
- Email is spam or not

5. 🧩 Ridge Regression

- A type of regularized regression that prevents **overfitting**
- Adds penalty to large coefficients

Example:

- When you have many features (like 100+ variables in a dataset)

6. 📊 Lasso Regression

- Similar to Ridge but can **remove irrelevant features** (feature selection)

Example:

- Selecting important factors affecting stock price

7. 🌐 Support Vector Regression (SVR)

- Based on Support Vector Machine
- Tries to fit data within a margin (tolerance)

Example:

- Predicting stock prices with tolerance for small errors

Quick Summary

Type	Use Case
Linear Regression	Simple straight-line prediction
Multiple Regression	Many inputs
Polynomial Regression	Curved patterns
Logistic Regression	Yes/No outcomes
Ridge/Lasso	Avoid overfitting
SVR	Flexible predictions



Linear Regression (Detailed Explanation)

Linear Regression is one of the simplest and most important algorithms in Machine Learning. It is used to model the relationship between **two variables** by fitting a **straight line**.



1. Basic Idea

Linear regression tries to answer:



“How does one variable change when another variable changes?”

- **Independent variable (X)** → input (e.g., experience)
- **Dependent variable (Y)** → output (e.g., salary)



2. Equation of Linear Regression

The equation of a straight line:

$$Y = mX + c$$

Where:

- **Y** = predicted value
- **X** = input variable
- **m** = slope (how much Y changes with X)
- **c** = intercept (value of Y when X = 0)



3. Simple Example



Problem:

Predict **salary** based on **years of experience**

Experience (X)	Salary (Y in ₹1000)
1	15
2	20
3	25
4	30

Step 1: Understand Pattern

- As experience increases → salary increases
- This looks like a straight-line relationship

Step 2: Fit Equation

From data, we can estimate:

$$Y = 5X + 10$$

Step 3: Make Prediction

If experience = 5 years:

$$Y = 5(5) + 10 = 35$$

👉 Predicted salary = ₹35,000

4. How It Finds the Best Line

Linear regression doesn't just guess the line—it uses a method called:

👉 Least Squares Method

Idea:

- Draw a line that minimizes the **error** between:
 - actual values
 - predicted values

Error = difference between real and predicted values



5. Important Terms

◆ Residual (Error)

$$\{\text{Error}\} = \{\text{Actual}\} - \{\text{Predicted}\}$$

◆ Cost Function

Measures total error:

Goal: **Minimize this error**

◆ Slope (m)

- Shows direction of relationship
- Positive slope → increasing trend
- Negative slope → decreasing trend



6. Graph Explanation (Concept)

- X-axis → input (experience)
- Y-axis → output (salary)
- Points → actual data
- Line → predicted values

The line is called the **best-fit line**



7. Assumptions of Linear Regression

For good results:

- Relationship should be **linear**
- No extreme outliers
- Data should be independent
- Constant variance (homoscedasticity)



8. Real-Life Examples

- Predict house prices
- Predict sales based on advertising
- Predict marks based on study hours
- Predict temperature trends